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Fibformer – a fibration transformer

Preprint · August 2025

DOI: 10.13140/RG.2.2.36293.10723

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Fibformer - a fibration transformer

Gary Nan Tie, Aug 22, 2025

Abstract We introduce a new transformer with novel fibration attention based on abductive reasoning. This coherent approach to query updating enables controlled structured exploration of continuation values.

Introduction

The generic transform attention mechanism (for sequence-to-sequence or next-token) applies a feed forward neural network element wise to query updates to introduce non-linearity in update refinement. In contrast, we learn a lifting neural network for a fibration, that takes into account the semantics of the key-value pairing by abductively choosing a key that explains a continuation value.

We adopt the notation from [1]:

queries $q_i \in Q$, $i \in [n]$

keys $k_j \in K$ and values $v_j \in V$, $j \in [m]$

$p: K \rightarrow V$ such that $p(k_j) =_E v_j$

$$\begin{array}{ccc}
 Q \otimes K & \xrightarrow{m \otimes 1} & Q' \otimes K \\
 1 \otimes p \downarrow & \nearrow \sigma & \downarrow 1 \otimes p \\
 Q \otimes V & \xrightarrow{m \otimes 1} & Q' \otimes V
 \end{array}$$

Learn σ so that this diagram commutes up to $=_E$, that is:

$$i) \quad \sigma \begin{bmatrix} q_i \\ p(k_j) \end{bmatrix} =_E \begin{bmatrix} q'_i \\ k_j \end{bmatrix}, \quad q'_i = m(q_i)$$

$$(+ \quad ii) \quad \pi_1 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} =_E q'_i$$

$$iii) \quad p \pi_2 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} =_E v_j$$

Query update step from q_i to $\tilde{q}_i$:

Learn neural network $\sigma: Q \otimes V \rightarrow Q' \otimes K$ from (†)

$$\text{with } q'_i \triangleq \sum_{j=1}^m \alpha_{ij} v_j,$$

where $\alpha_{ij} = \text{softmax score}(q_i, k_j)$.

Let $j^* = \arg \max_j \alpha_{ij}$.

Define update of q_i to be: $\tilde{q}_i \triangleq \pi_1 \sigma \begin{bmatrix} q'_i \\ v_{j^*} \end{bmatrix}$

To refine query q_i , replace q_i with $\tilde{q}_i$, and successively update, until perplexity is acceptable.

Instead of applying a Feed Forward neural network to q'_i , as in generic transformer attention, we learn a lifting neural network σ , for fibration $1 \otimes P$.

This abductive reasoning takes into account the key-value pairing semantics (see references).

References

[1] 'FibChat - It makes stuff up, but with structure.'

Gary Nan Tie · July 2025

DOI: 10.13140/RG.2.2.10549.59364/2

FibChat is a LLM based on fibration attention, that enables coherent structured mitigation of hallucinations.

[2] 'Fibrations explain all you need!

- Fibrations, Abduction, Attention

Gary Nan Tie, Mar 4, 2025

DOI: 10.13140/RG.2.2.35984.52488

To conceptualize and perform abductive machine learning, categorical fibrations are proposed, as they coherently give both context and structure for abductive reasoning via an attention mechanism. Moreover, 2-categorical lifting compatibility conditions ensure consistent hierarchical and parallel explanations.

[3] 'Neural Network Abduction'

Gary Nan Tie, Jun 13, 2025

DOI: 10.13140/RG.2.2.18506.07360/1

Abductive reasoning has a fibration semantics that can be implemented by neural networks; a step towards artificial general intelligence.

[4] 'Parsimonious Neural Network Abduction'

Gary Nan Tie, Jun22, 2025

DOI: 10.13140/RG.2.2.15695.80804

For hypotheses whose effect is manifested by observations, abduction seeks to explain a given observation by finding a hypothesis whose effect is that observation. Abductive reasoning has a fibration semantics that can be implemented by neural networks. In this note we introduce a parsimonious choice of explanation according to one's utility function.

[5] 'Attention Fibration Abduction'

Gary Nan Tie, Jul 2, 2025

DOI: 10.13140/RG.2.2.29060.23680/1

Transformer attention and abductive reasoning are both instances of a weak factorization system fibration, implemented by neural networks.